

COMPASS_Synoptic_LE_Soil-Fluxes_2024_CH4

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0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data
 - a. i.e. drops fluxes that have the drop column TRUE or FLAG this
 - b. adjusting times as needed
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Label format

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
[1] "CSV file exists and has been read into the code."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

Warning: There were 6 warnings in `mutate()`.

The first warning was:

```
i In argument: `drop = letter_order < max(letter_order, na.rm = TRUE)`.
```

```
i In group 8: `core_id = "2024_02_02_CRC_UP_25"`.
```

```
Caused by warning in `max()`:  
! no non-missing arguments to max; returning -Inf  
i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

0.8 fixing some Deadbands Revisit here, it is not working to fix the dead band

##Extracting initial CH4 concentration

0.9 Change the units from ppb to nmol

```
#change the units  
alldat_trimmed$CH4_nmol <- ffi_ppb_to_nmol(alldat_trimmed$ch4,  
                                           volume = (alldat_trimmed$TotalVolume_fixed_cm3 / 1000000), #  
                                           temp = alldat_trimmed$chamber_t)      # degrees C
```

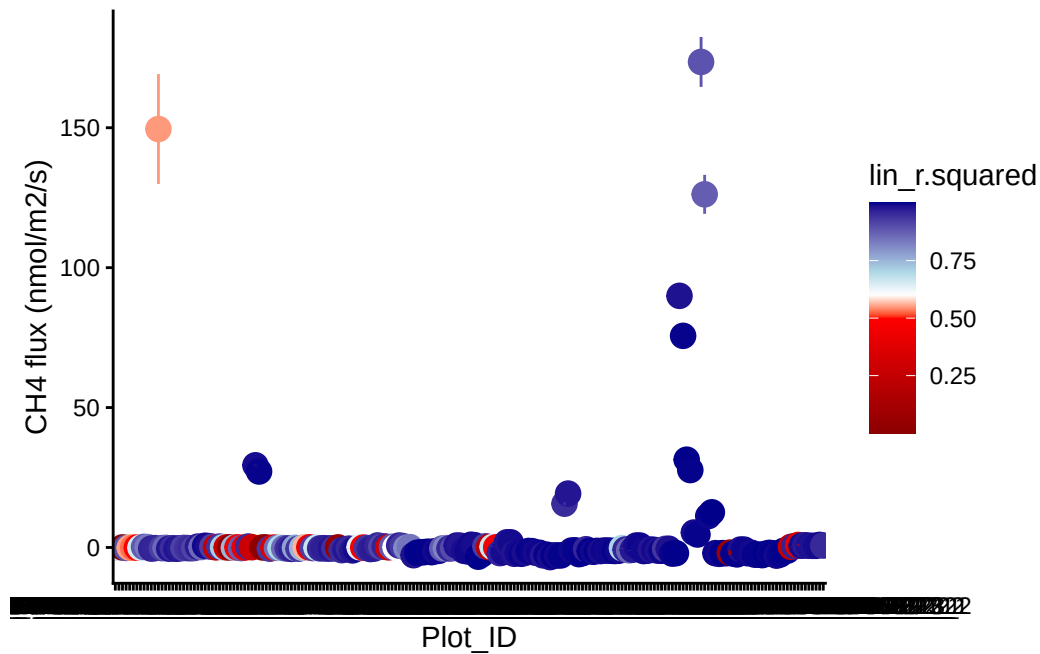
Assuming atm = 101325 Pa

Using R = 8.31446261815324 m3 Pa K-1 mol-1

```
#> Assuming atm = 101325 Pa  
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1  
  
alldat_trimmed$CH4_nmol_m2 <- alldat_trimmed$CH4_nmol / 0.0318 #adjust for the area
```

0.10 Calculate fluxes

0.11 Visualize fluxes and sort by r2



0.12 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  group_by(Plot_ID) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP),
    .groups = "drop"
  )

# use full alldat + trim limits to classify each point
r_all <- alldat %>%
  filter(Plot_ID %in% time_overrides$Plot_ID) %>%
  left_join(trim_limits, by = "Plot_ID") %>%
```

```

group_by(Plot_ID) %>%
mutate(
  time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
  point_status = case_when(
    TIMESTAMP < t_start_trim ~ "Discarded (DeadBand)",
    TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
    TRUE                     ~ "Used"
  )
) %>%
ungroup()

# ---- PLOTTING: one plot per Plot_ID (PDF friendly) ----

plot_ids <- unique(r_all$Plot_ID)

for (P in plot_ids) {

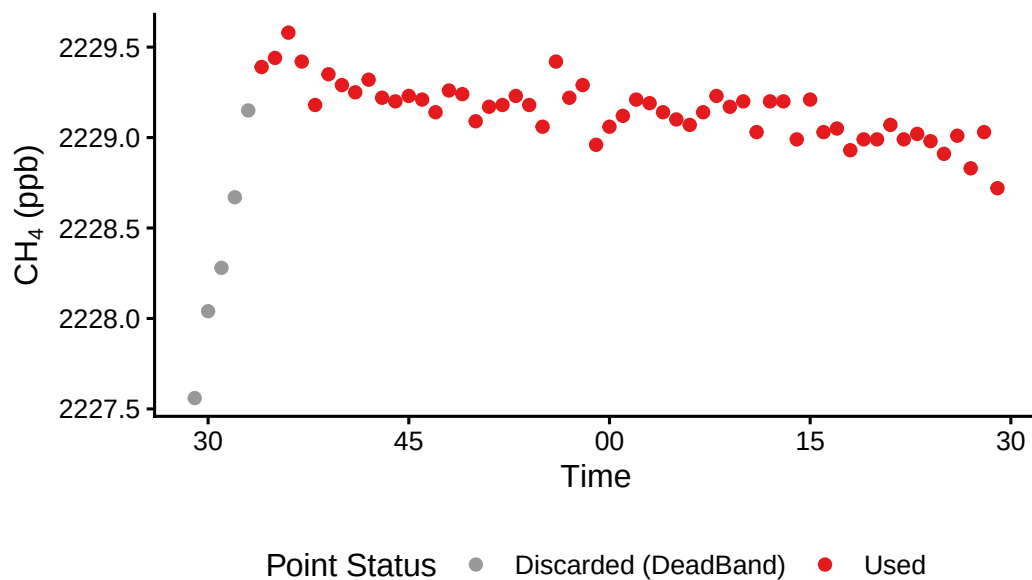
  p <- ggplot(
    r_all %>% filter(Plot_ID == P),
    aes(x = TIMESTAMP, y = ch4, color = point_status)
  ) +
    geom_point(size = 1.8) +
    scale_color_manual(
      values = c(
        "Used" = "#E41A1C",
        "Discarded (DeadBand)" = "grey60",
        "Discarded (stop trim)" = "grey20"
      )
    ) +
    theme_classic(base_size = 12) +
    ylab(expression(paste(CH[4], " (ppb)"))) +
    xlab("Time") +
    labs(
      title = paste("Incubation Period Adjustments:", P),
      color = "Point Status"
    ) +
    theme(
      legend.position = "bottom",
      plot.title = element_text(face = "bold")
    )

  print(p)  # IMPORTANT: each print() = new page in PDF
}

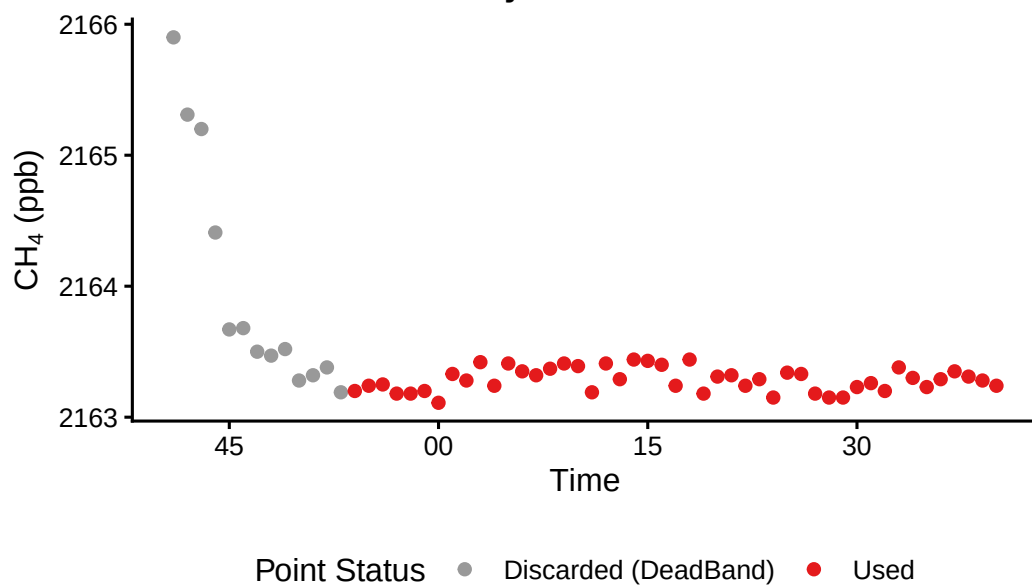
```

}

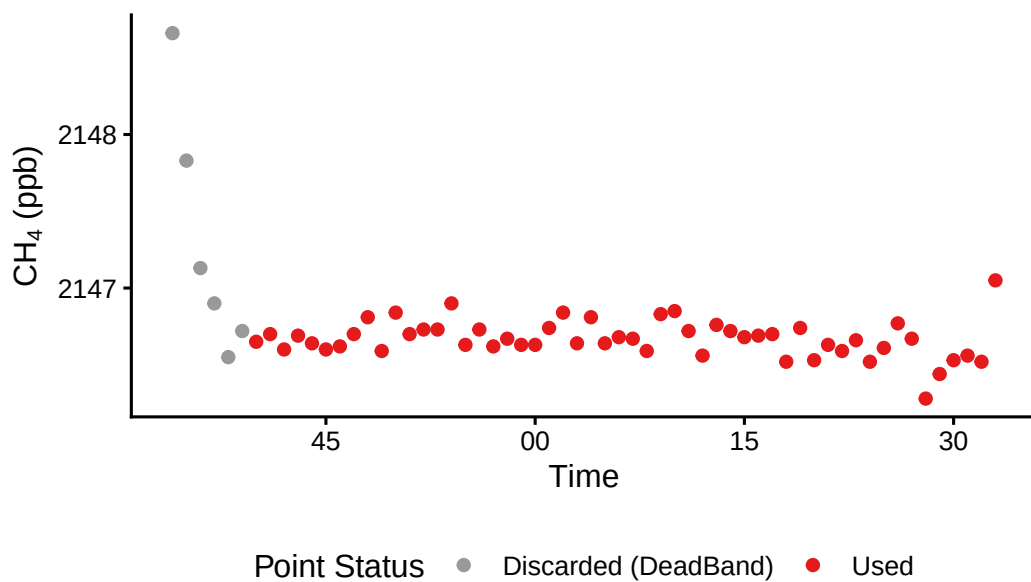
Incubation Period Adjustments: 2024_02_23_OWC_



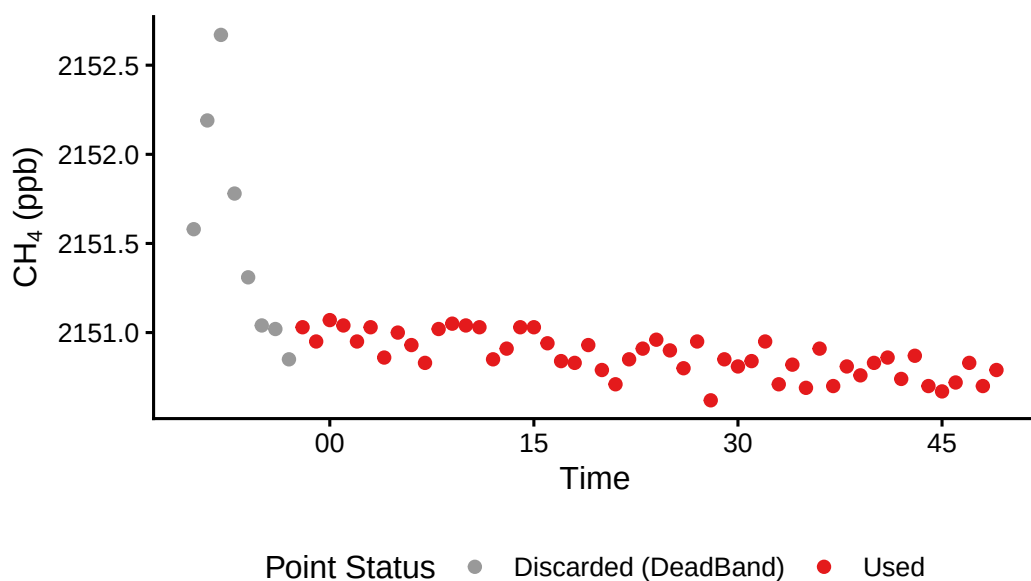
Incubation Period Adjustments: 2024_02_23_OWC_ \



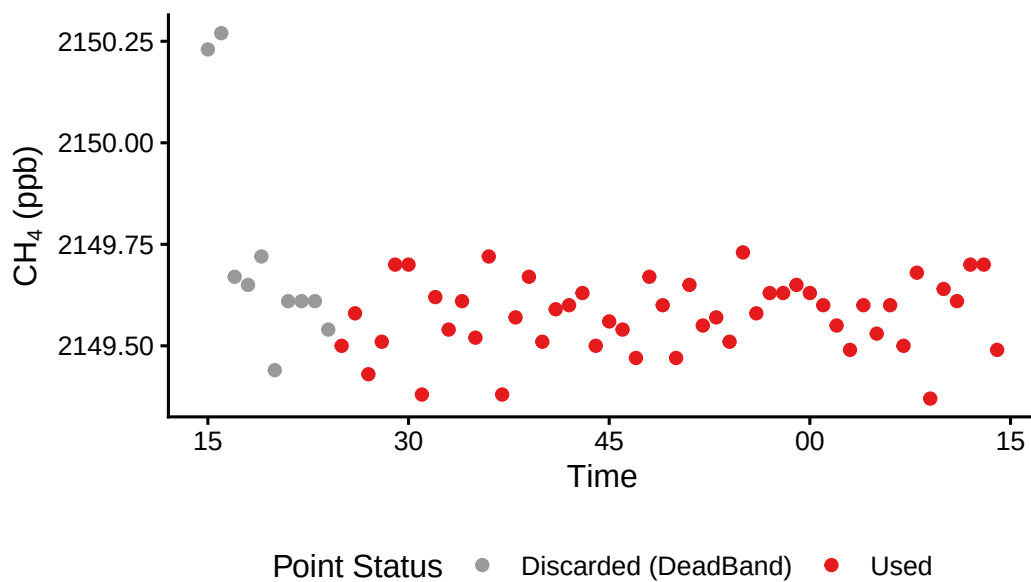
Incubation Period Adjustments: 2024_02_23_OWC_1



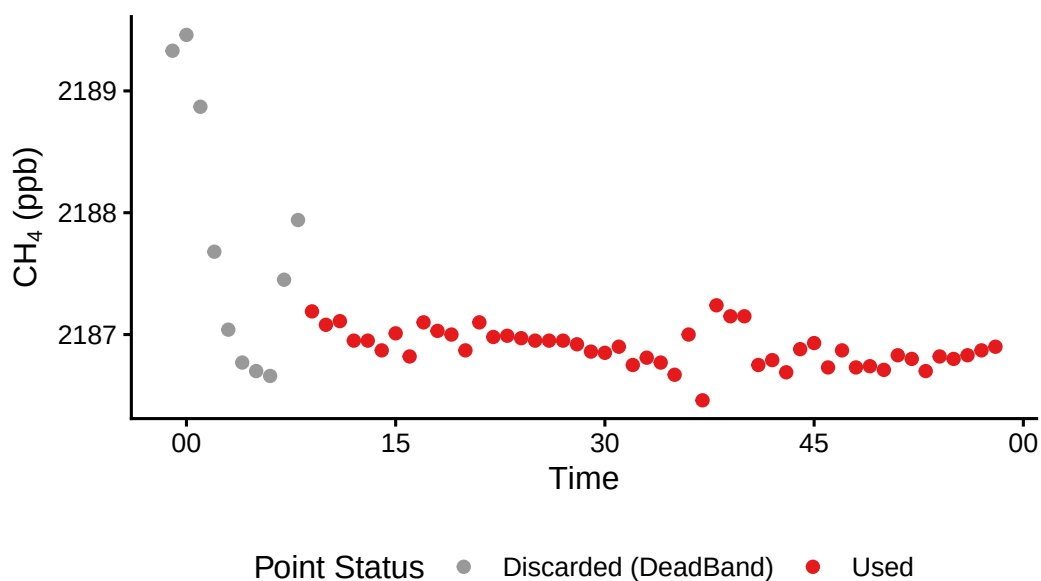
Incubation Period Adjustments: 2024_02_23_OWC_



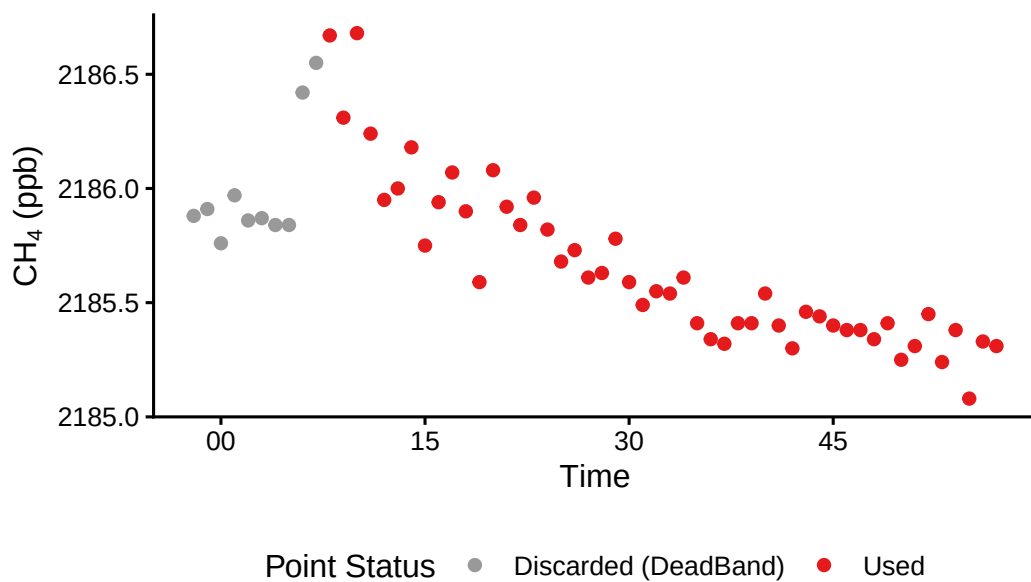
Incubation Period Adjustments: 2024_02_23_OWC



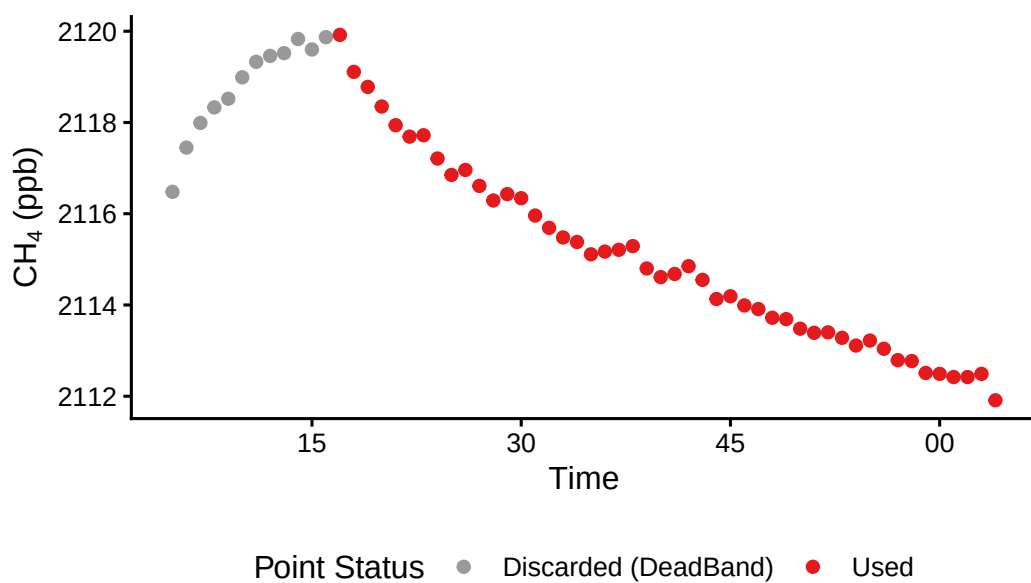
Incubation Period Adjustments: 2024_02_23_OWC_1



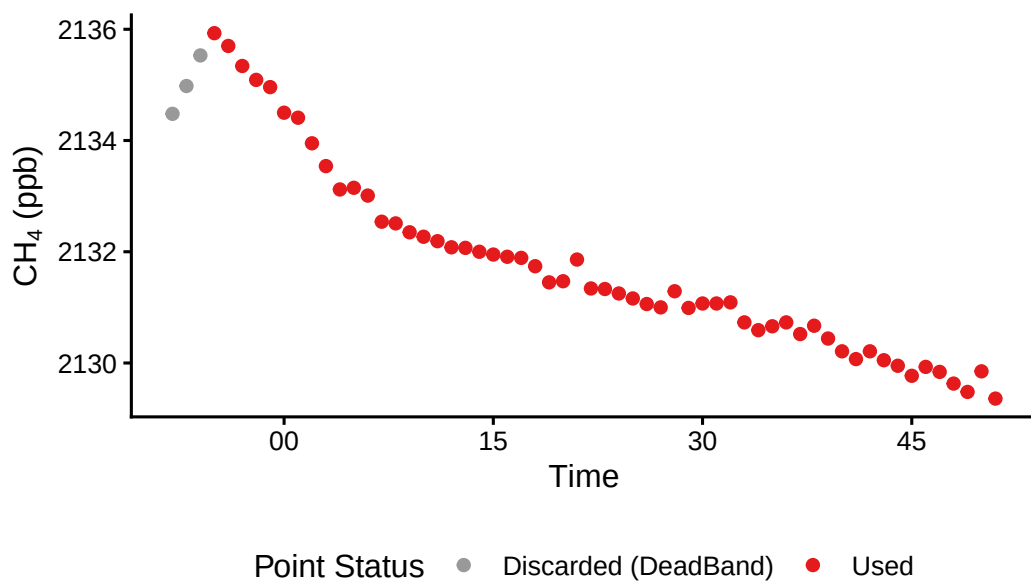
Incubation Period Adjustments: 2024_02_23_OWC_



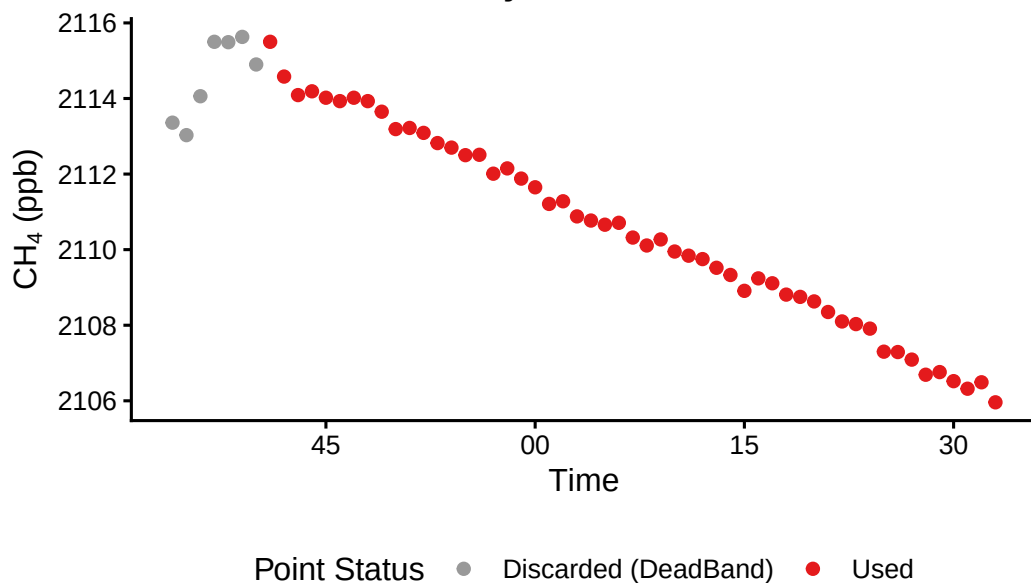
Incubation Period Adjustments: 2024_07_31_OWC_1



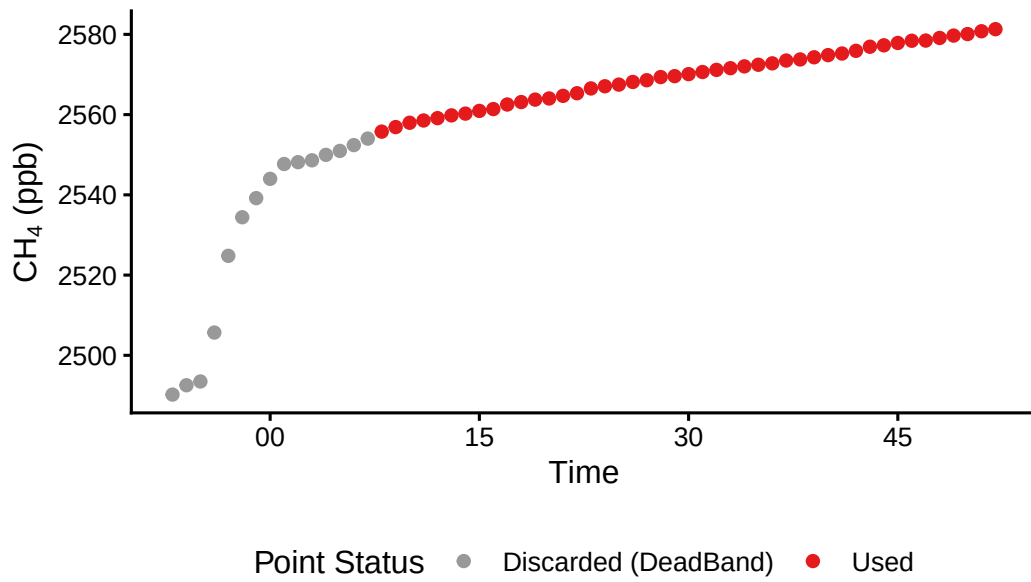
Incubation Period Adjustments: 2024_07_31_OWC_I



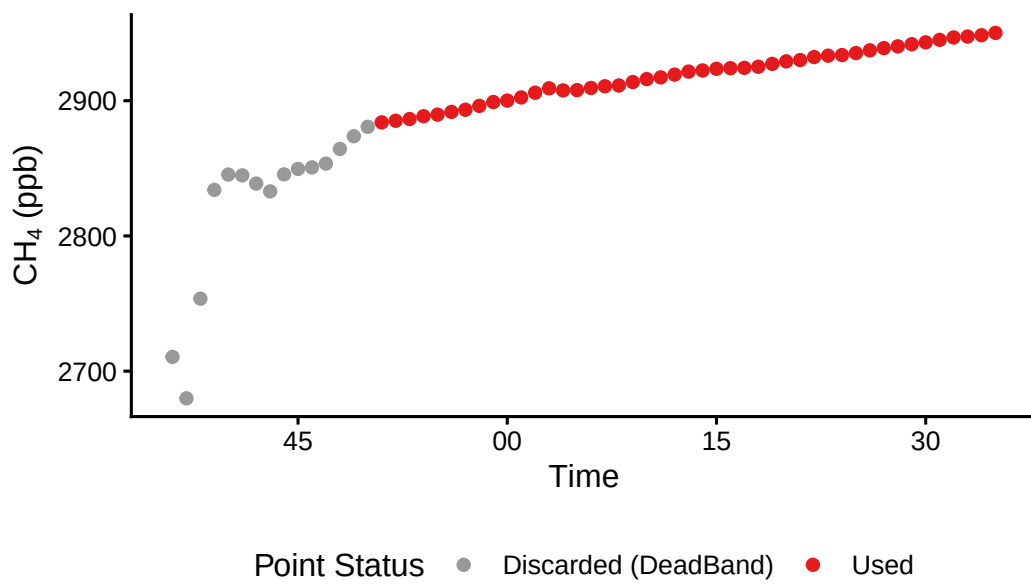
Incubation Period Adjustments: 2024_07_31_OWC_I

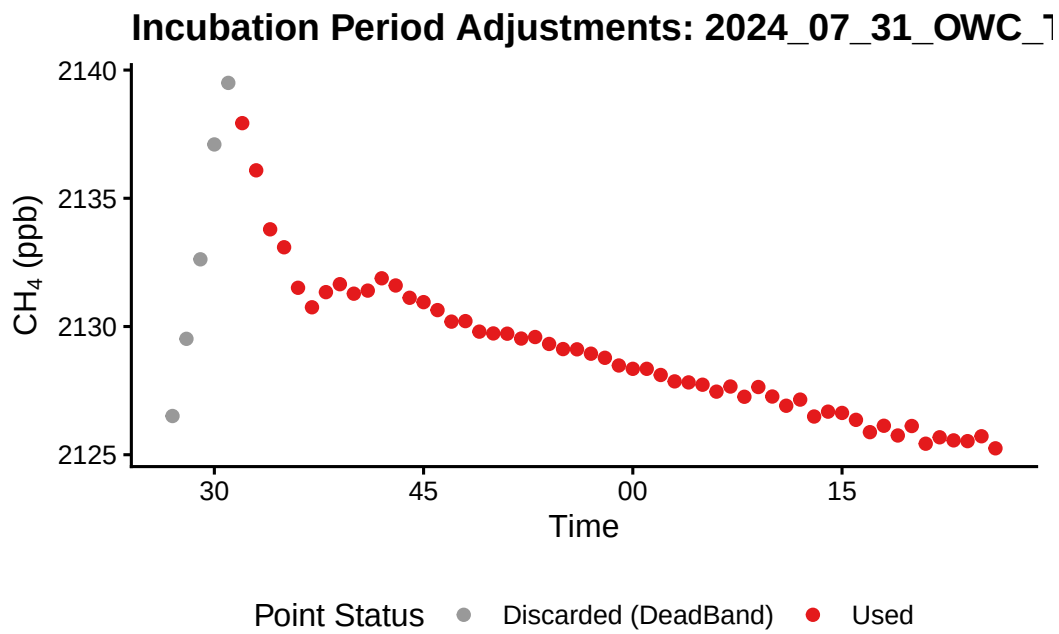
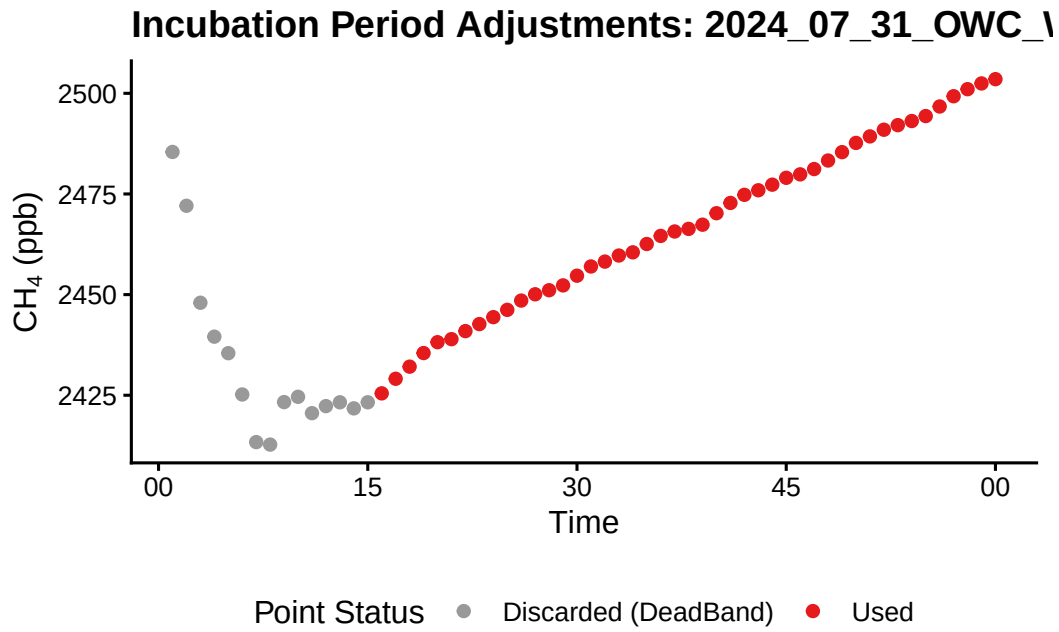


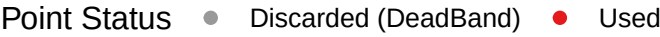
Incubation Period Adjustments: 2024_07_31_OWC_1



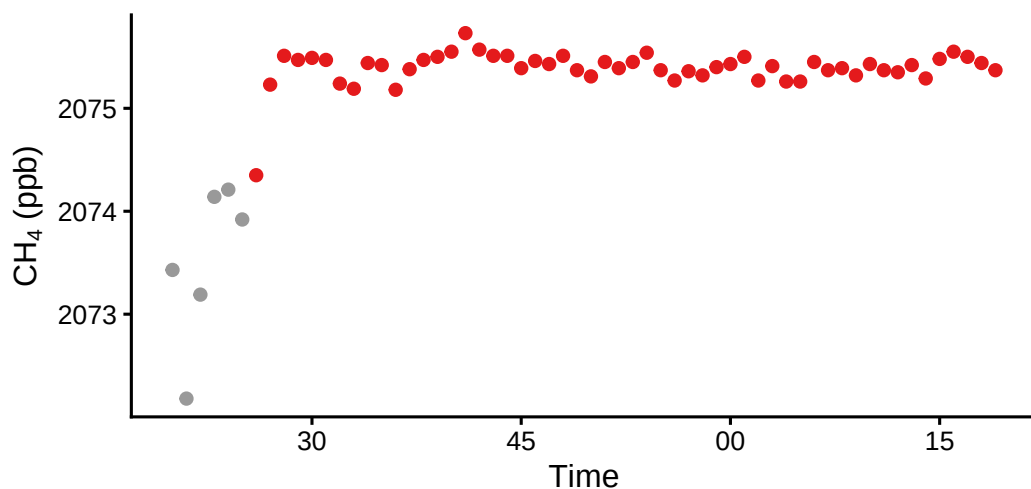
Incubation Period Adjustments: 2024_07_31_OWC_1



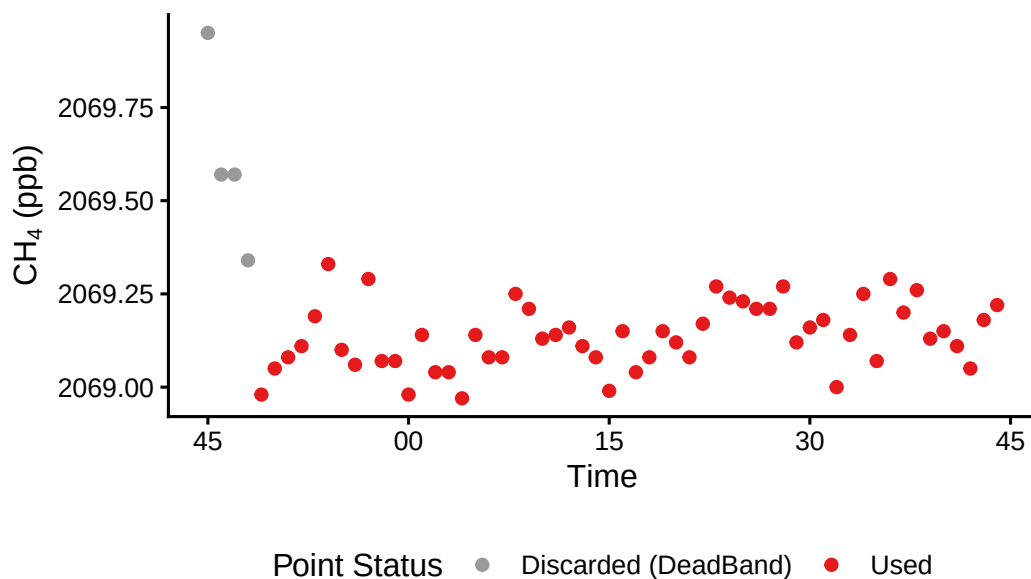




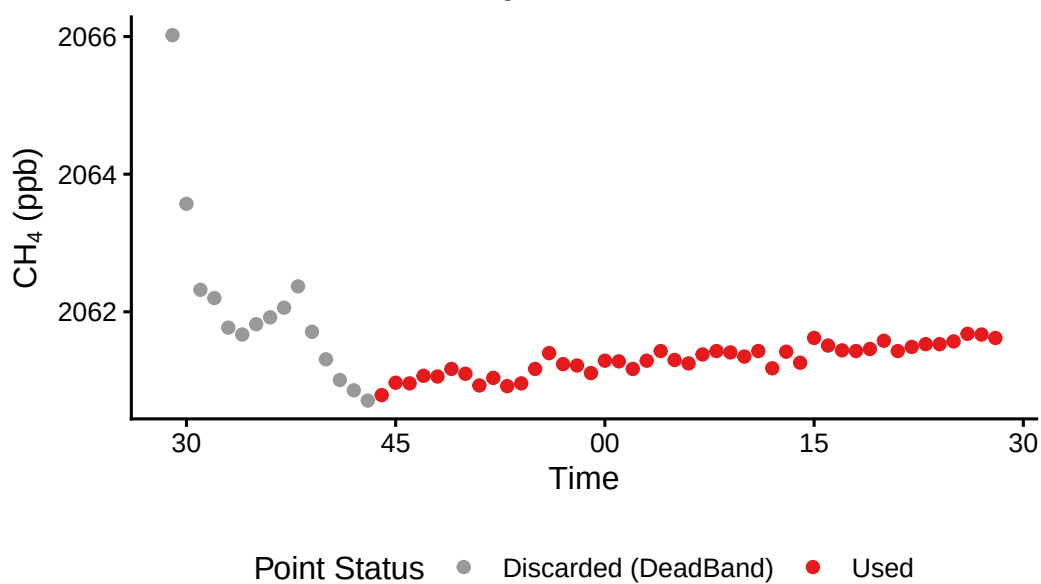
Incubation Period Adjustments: 2024_02_02_CRC_T



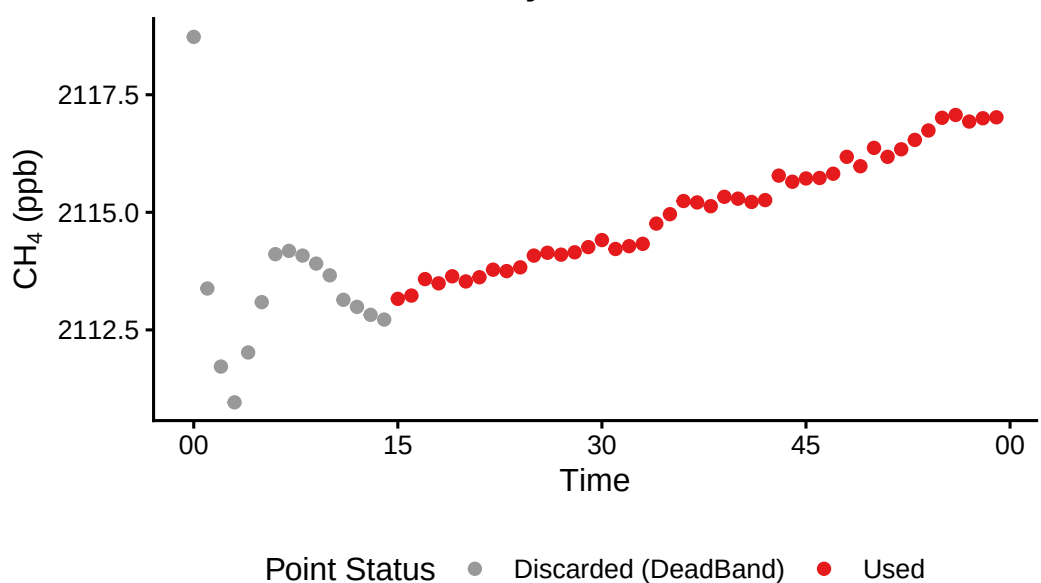
Incubation Period Adjustments: 2024_02_02_CRC



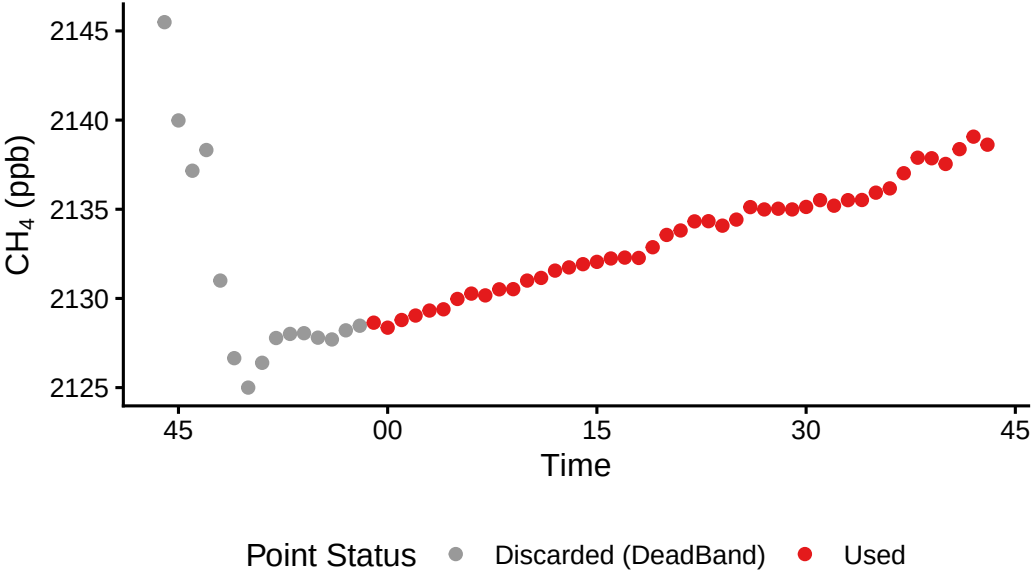
Incubation Period Adjustments: 2024_02_02_CRC_V



Incubation Period Adjustments: 2024_07_30_CRC_

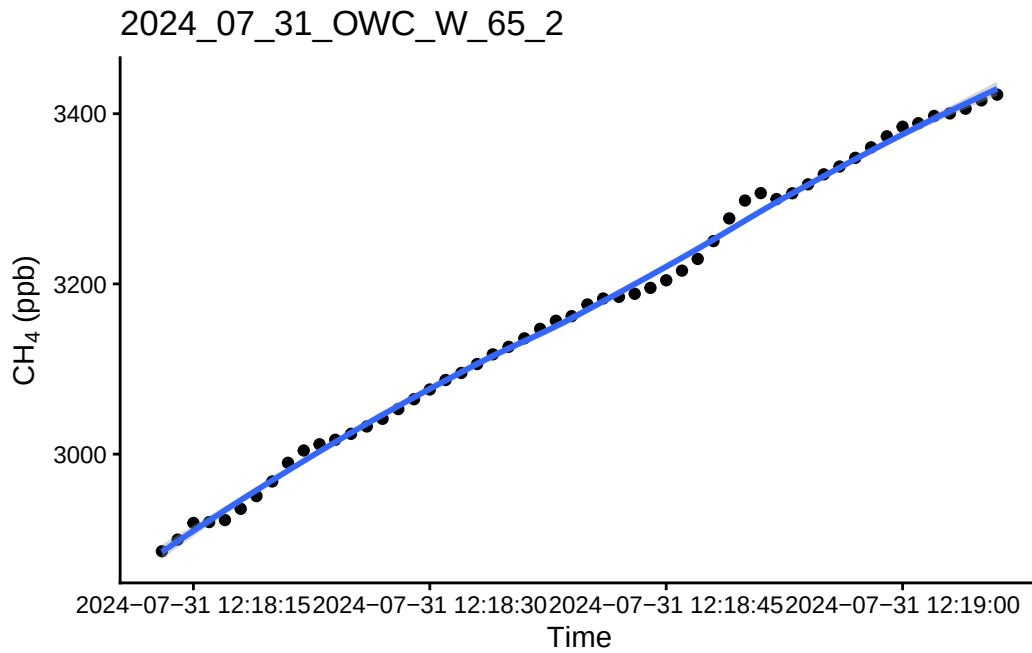


Incubation Period Adjustments: 2024_07_30_CRC_L

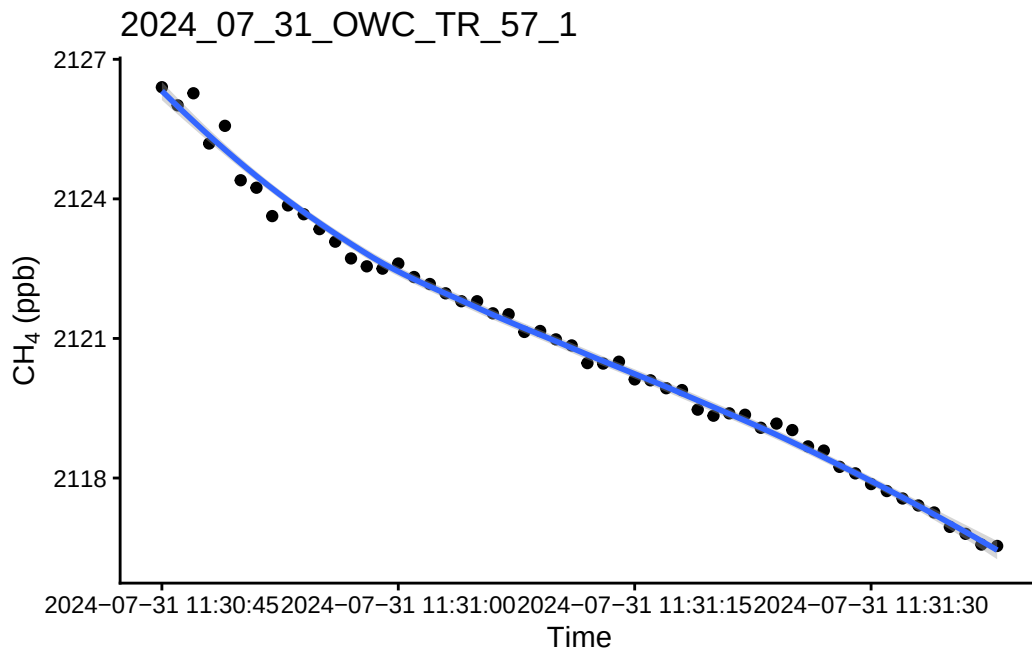


0.13 Spot check a few fluxes with $R^2 > 0.9$

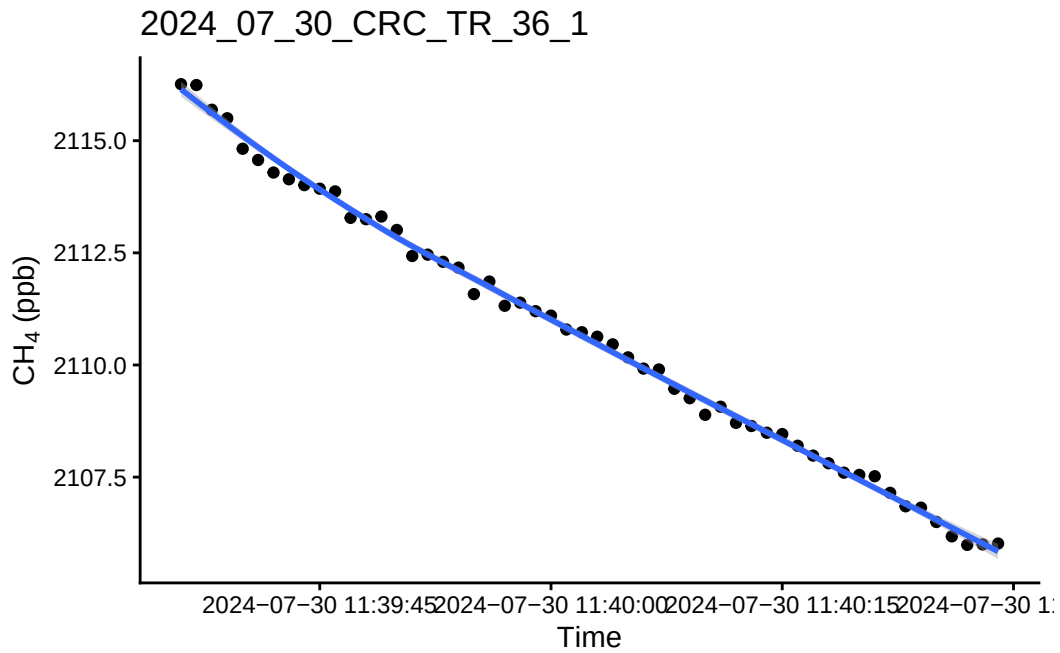
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



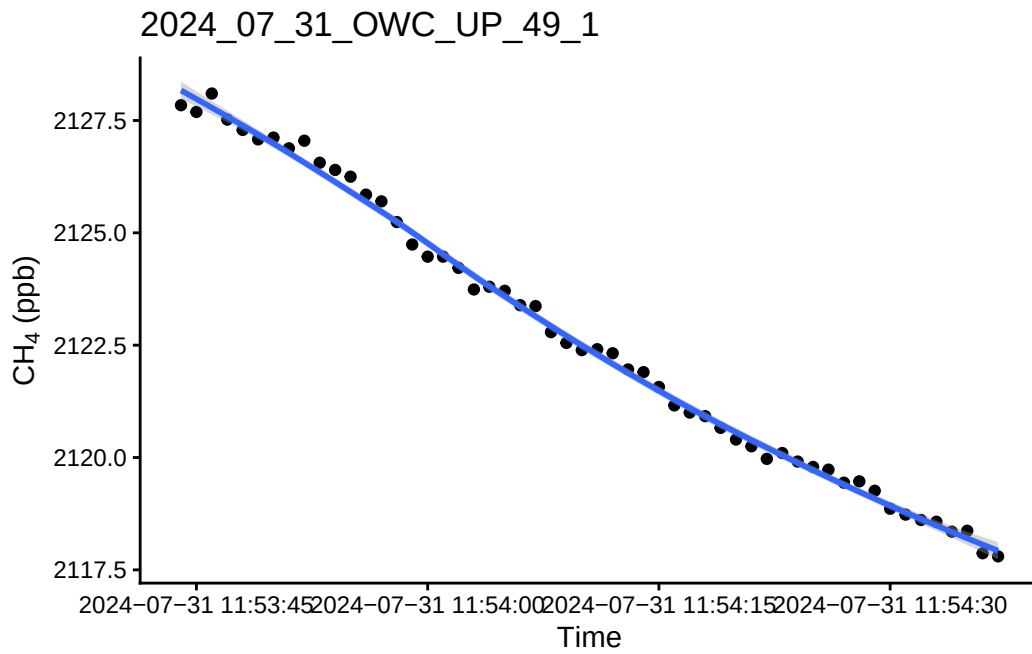
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



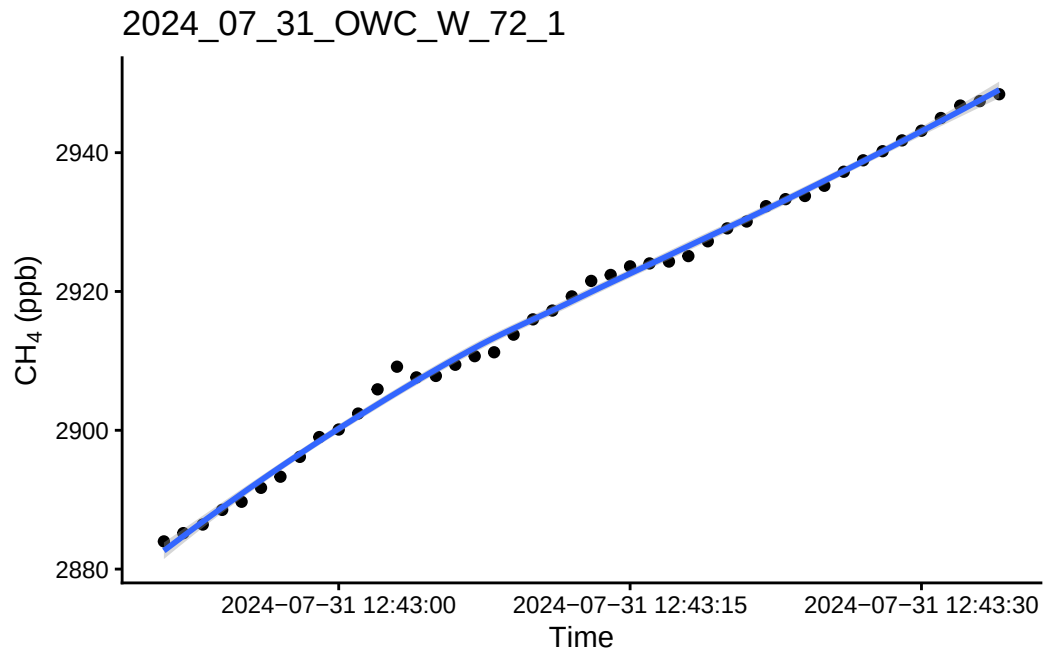
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

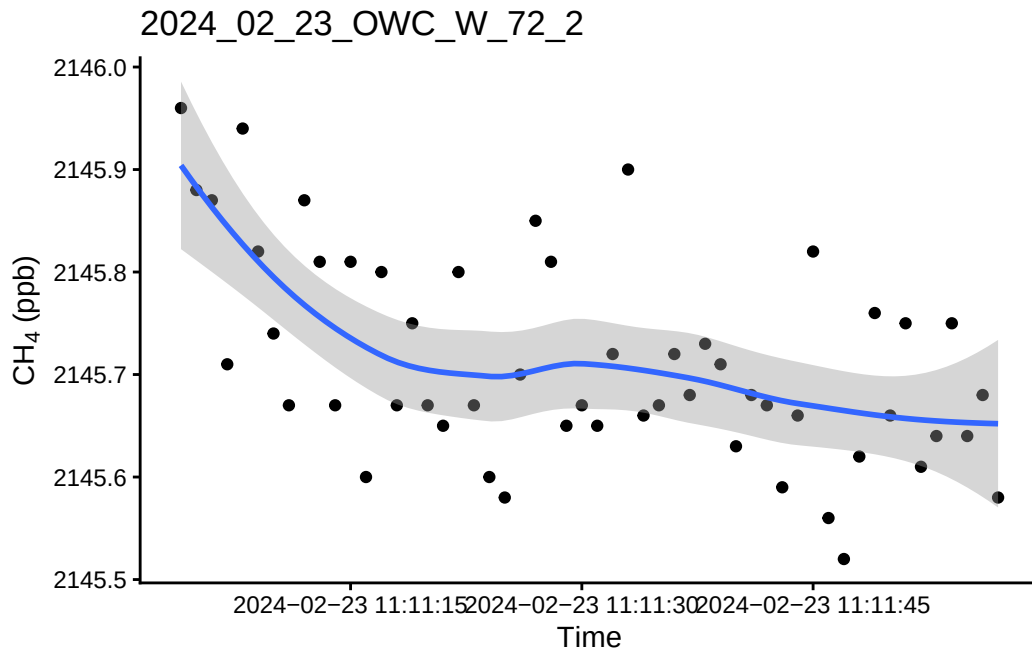


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

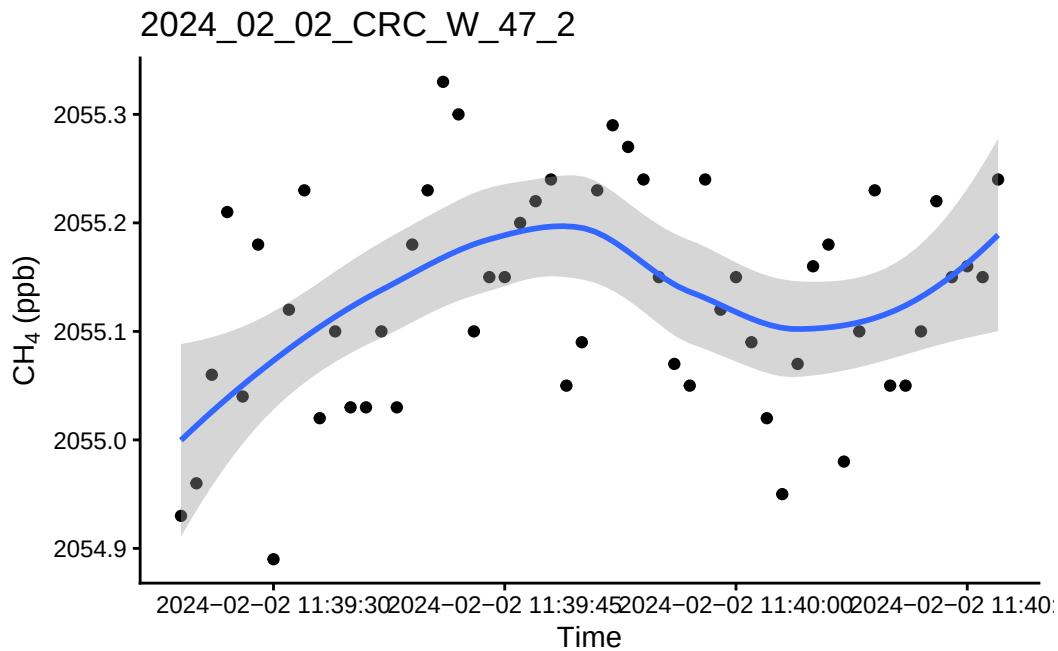


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

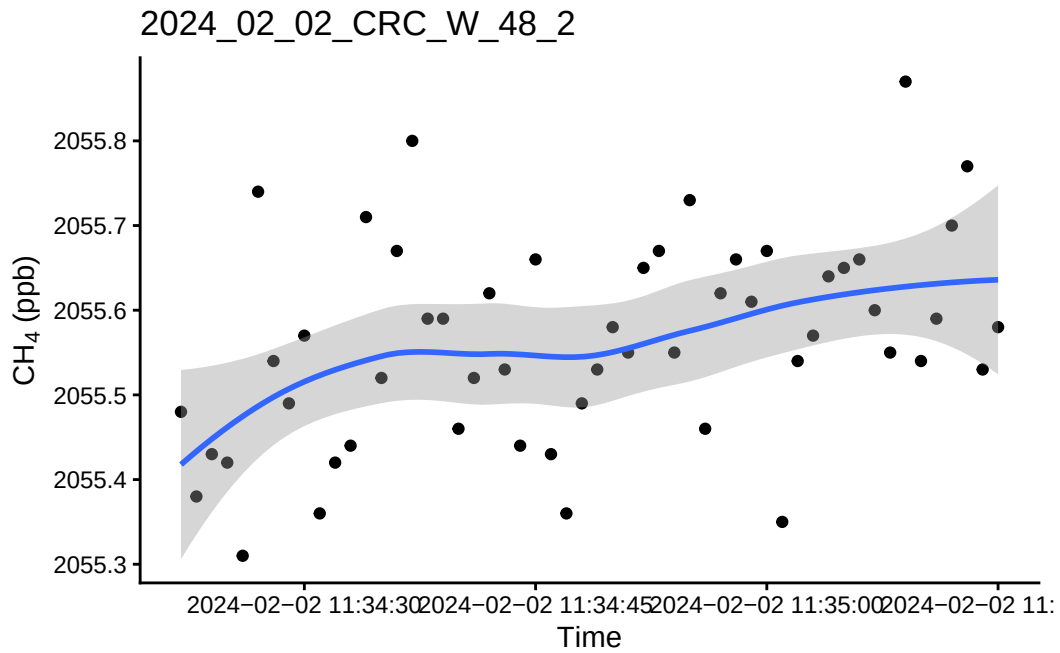
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



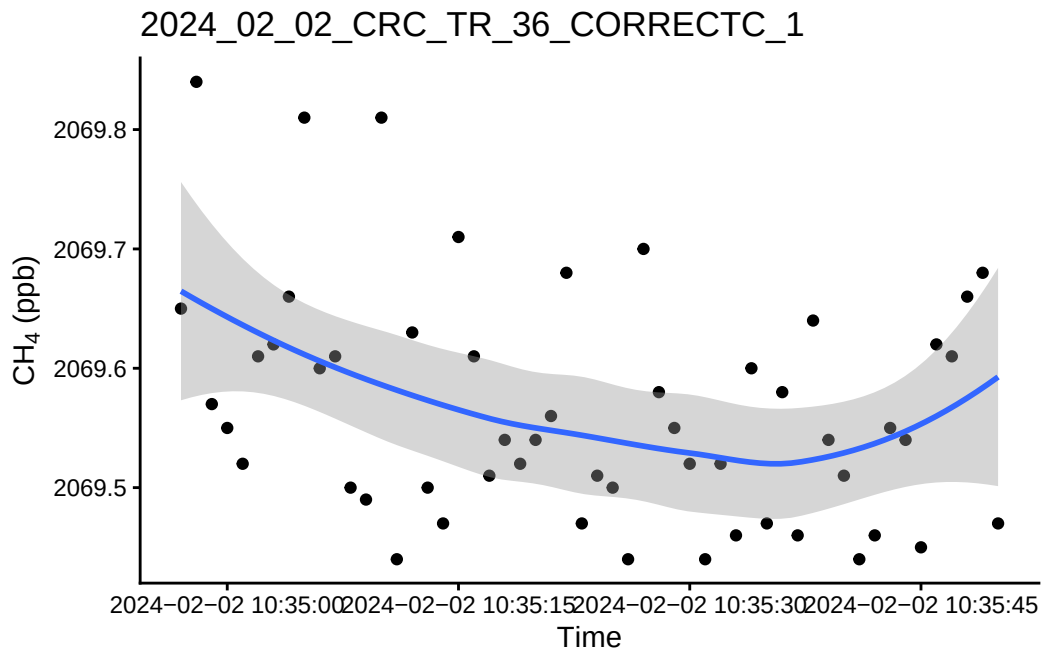
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



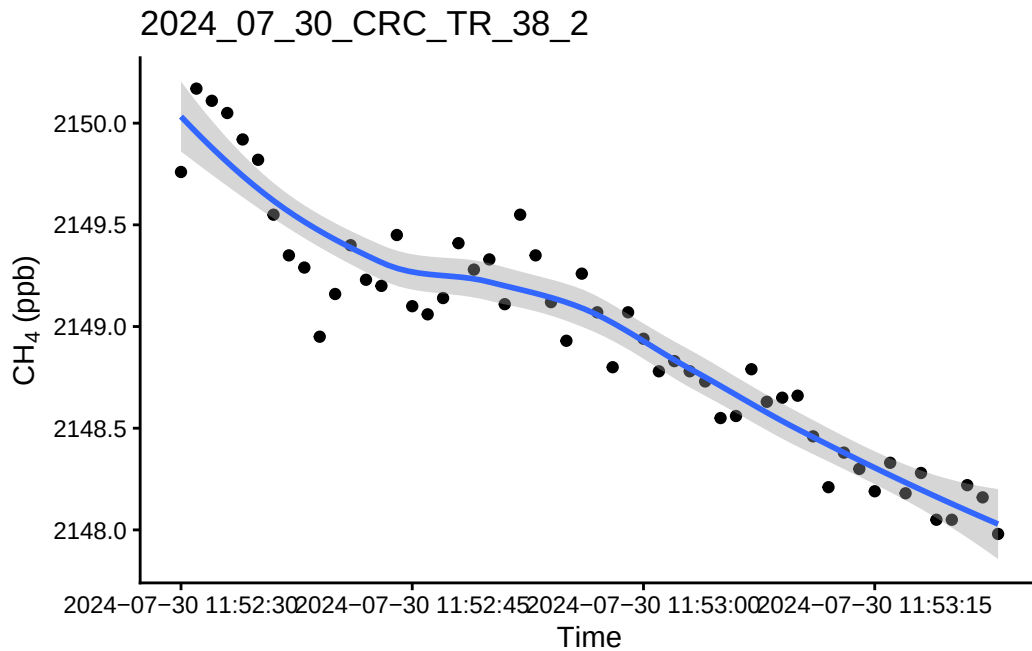
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



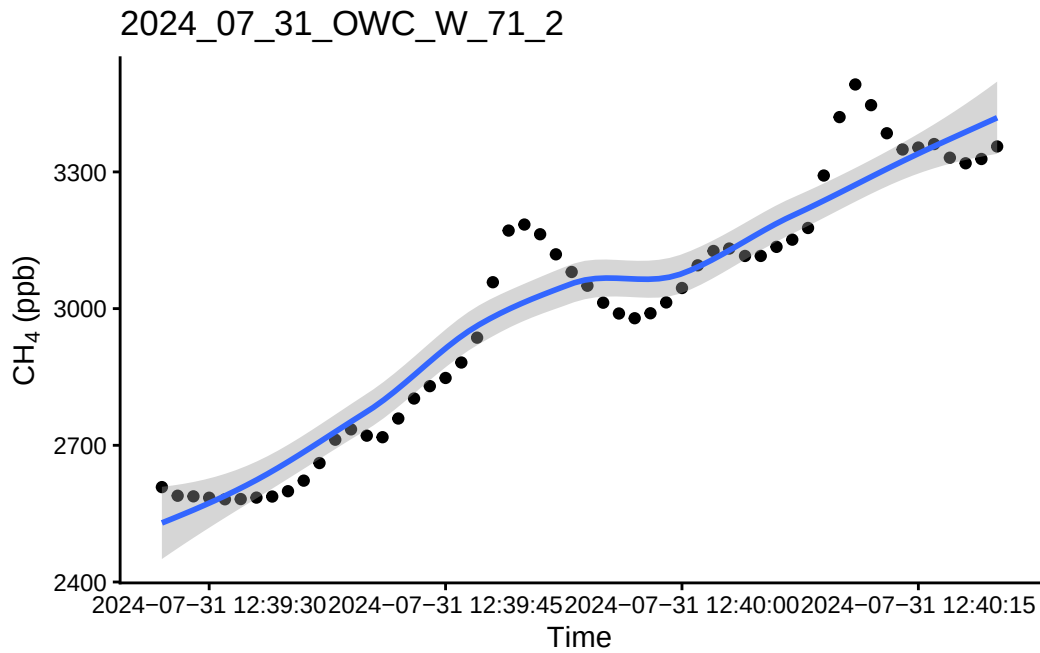
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



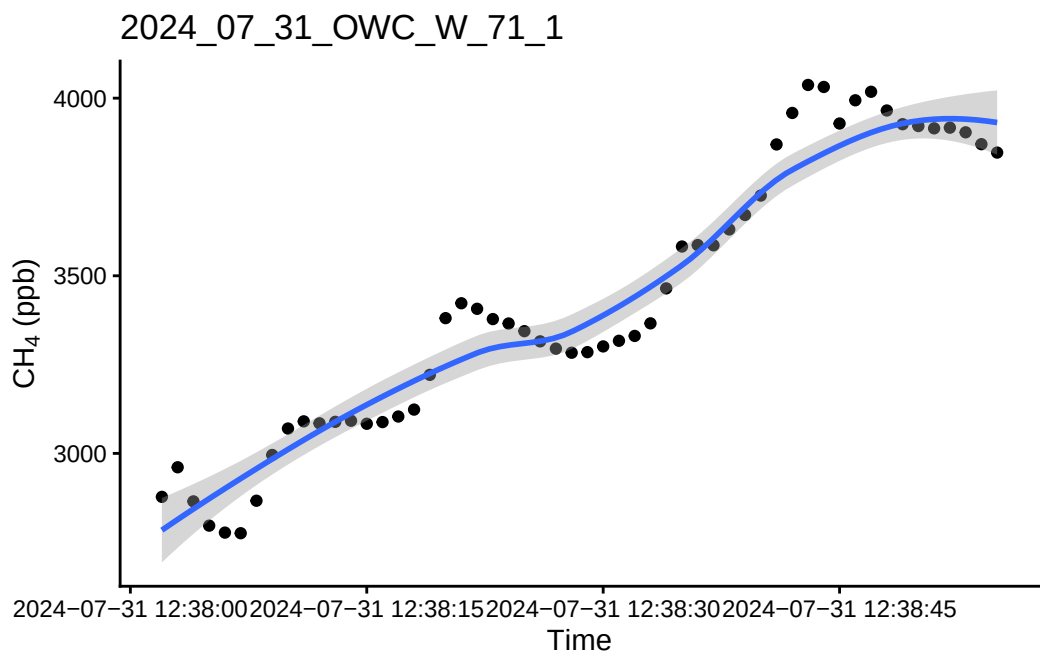
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

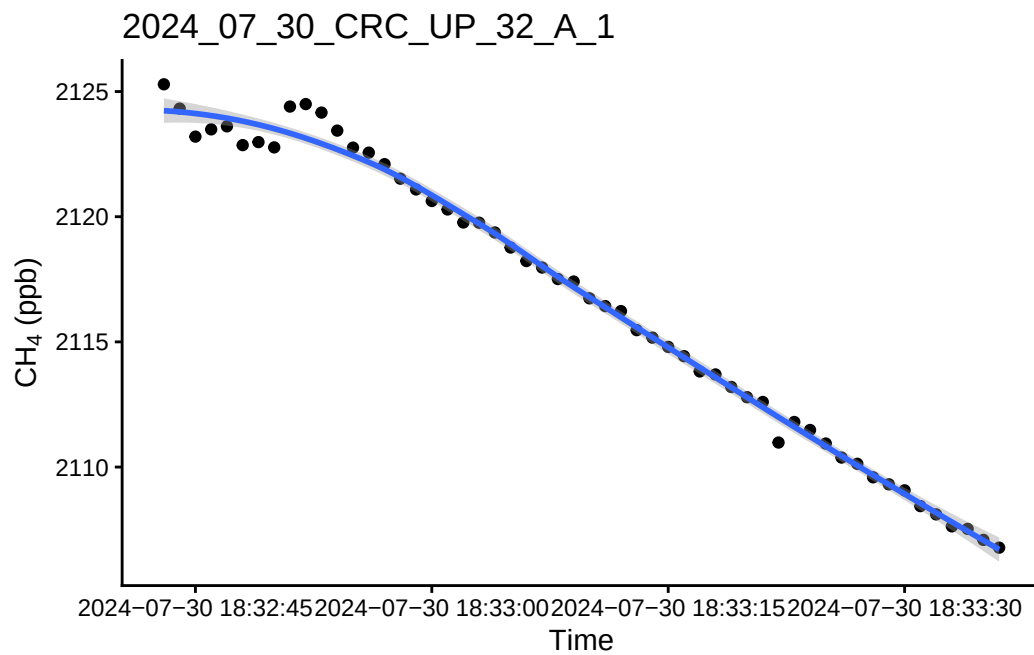


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

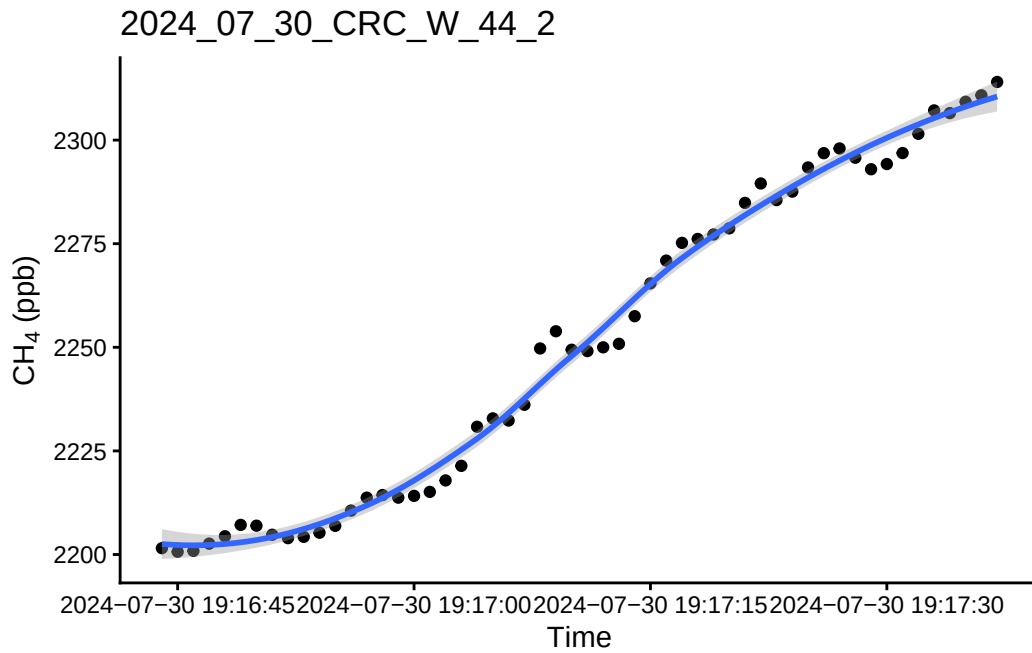


1.1 Look at the really high or really low ones to see if there is a timing issue or something

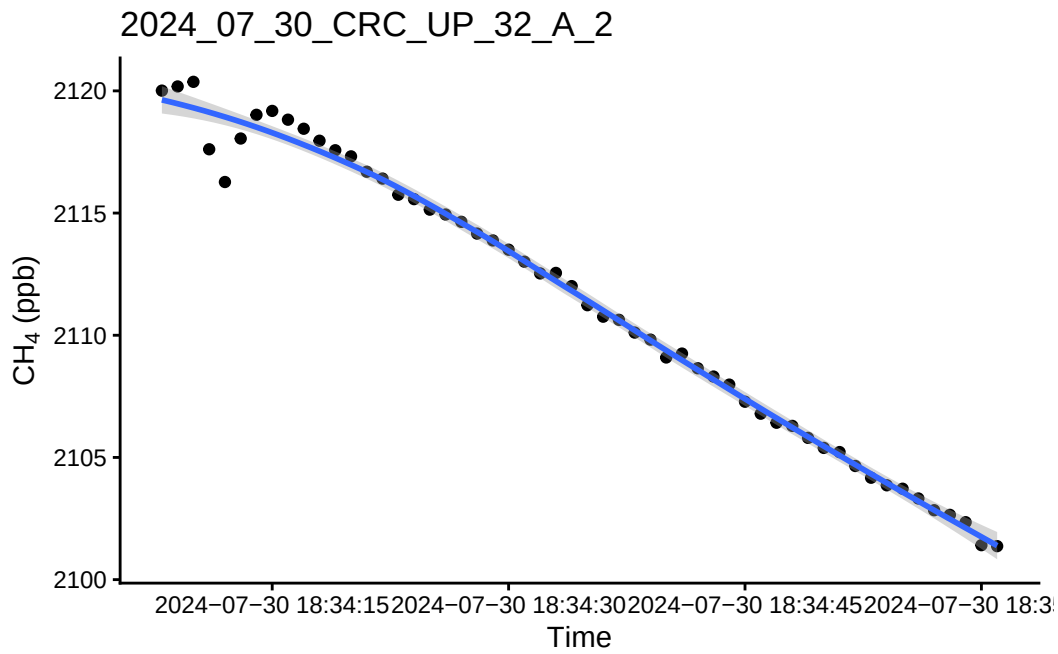
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



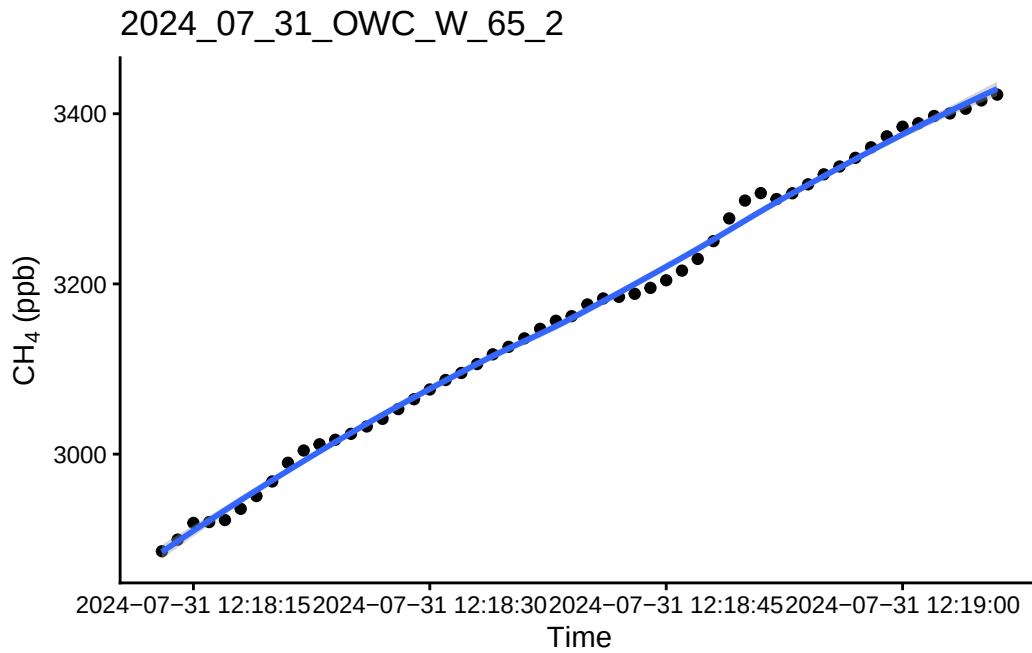
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



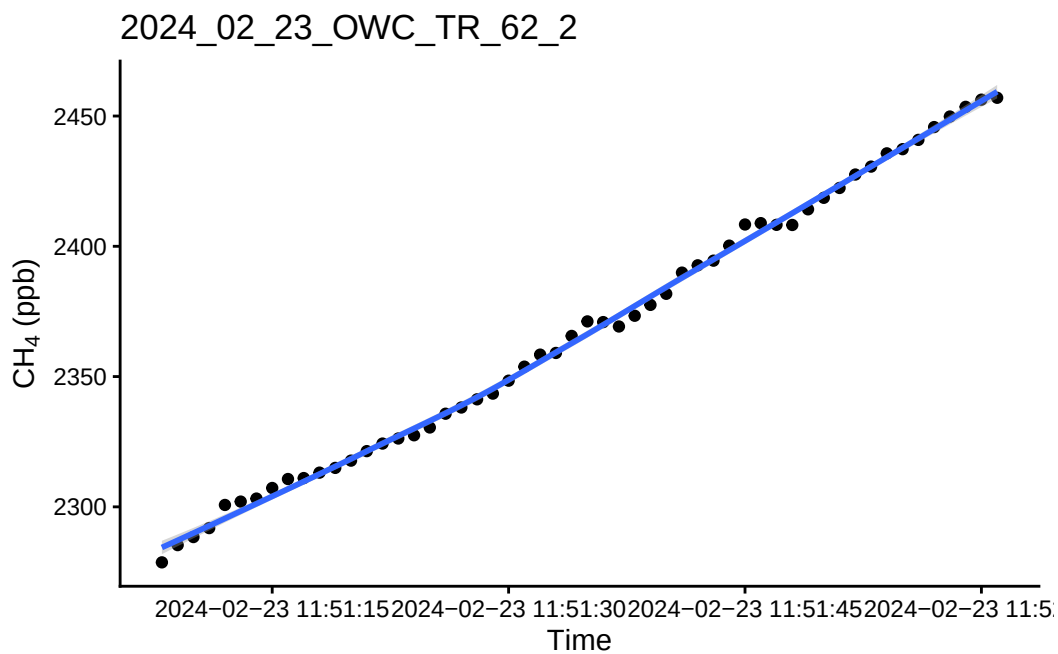
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

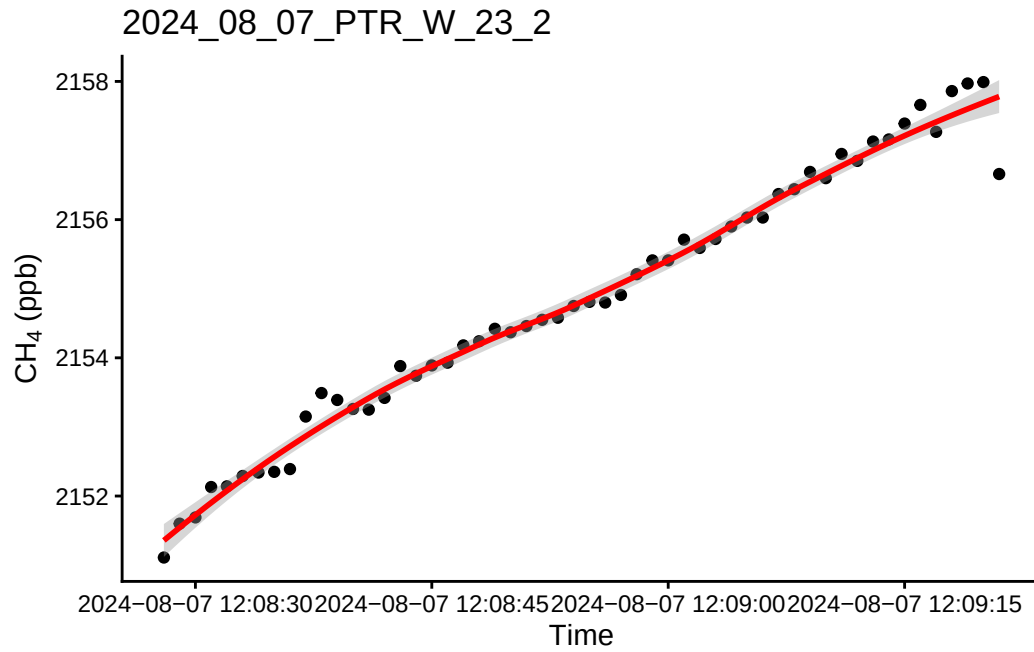


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

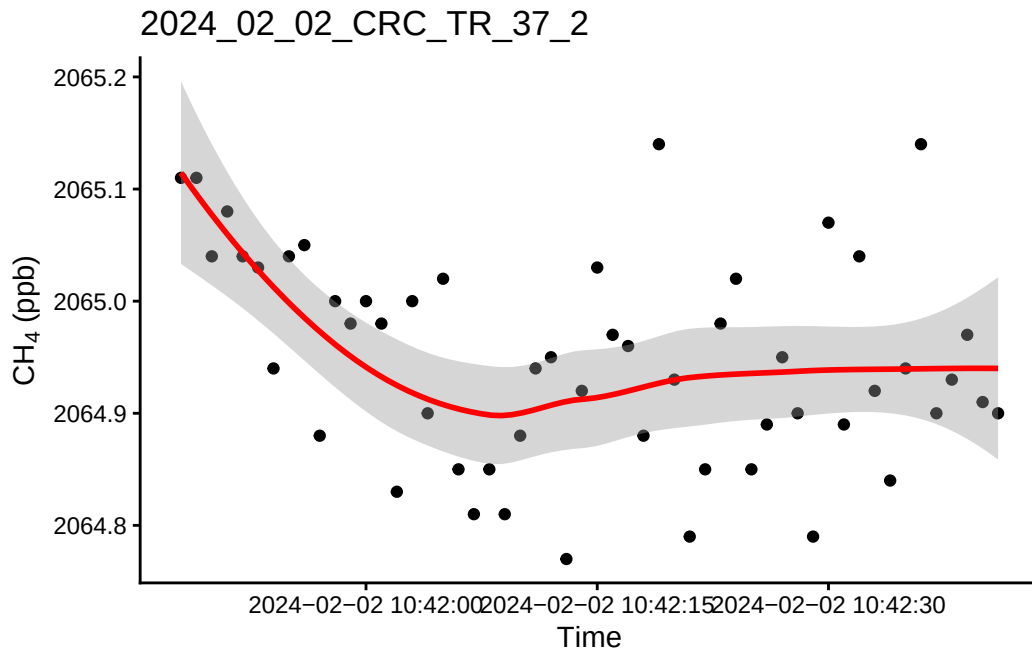


1.2 Check all fluxes

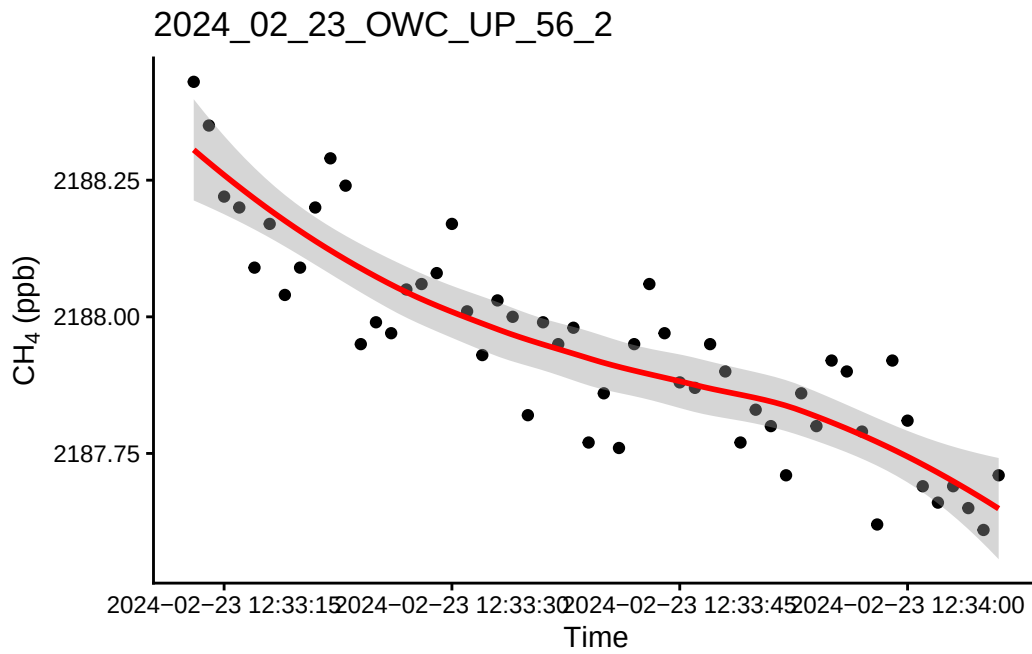
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



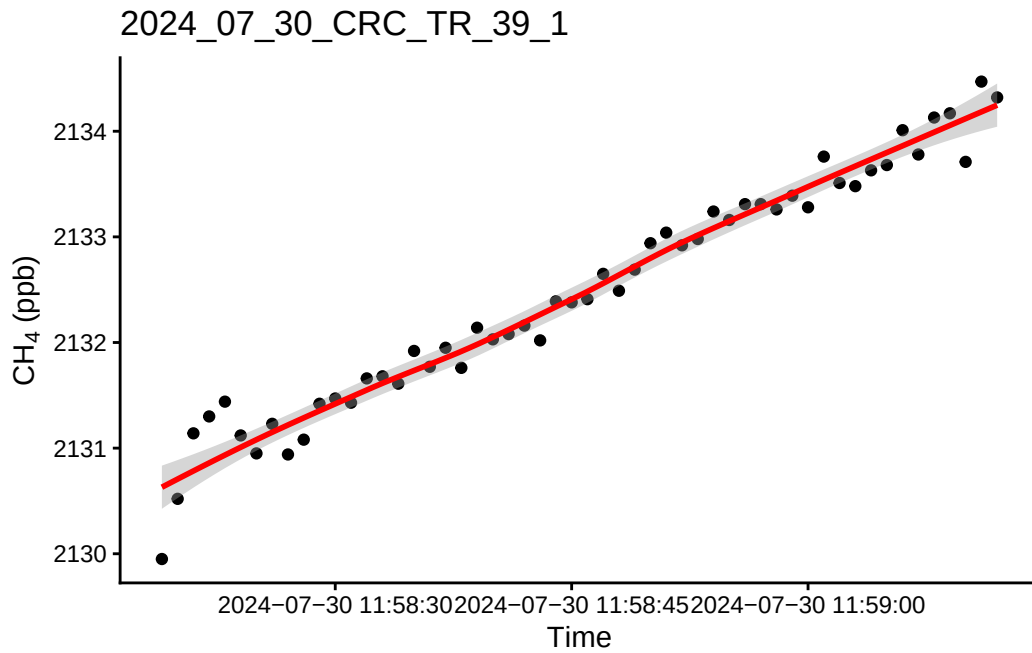
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



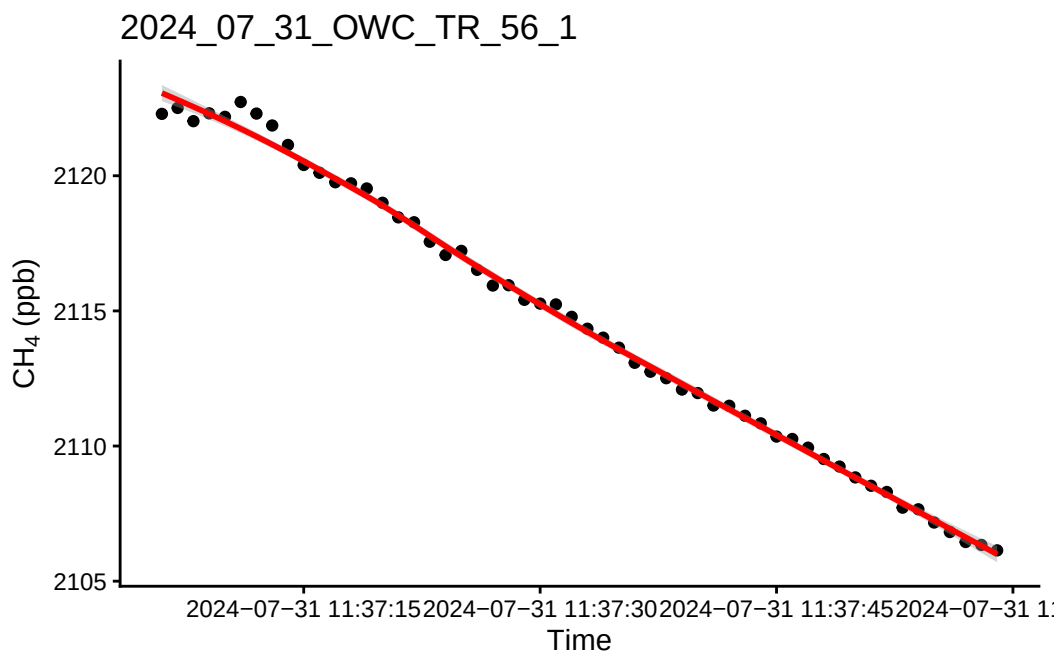
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



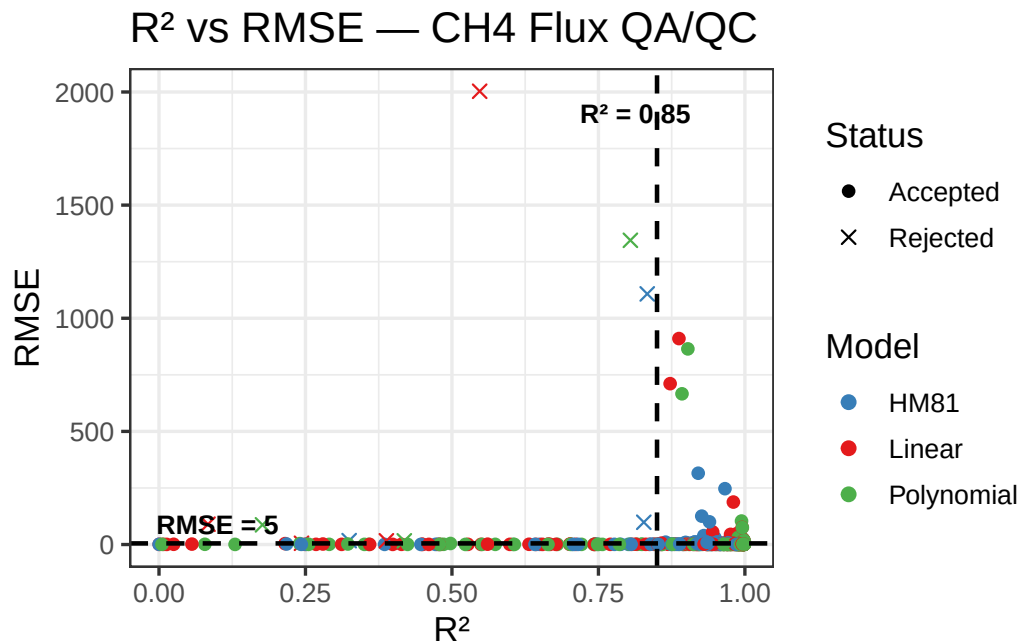
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



1.3 QAQC - Flags

Table 1: QA/QC CH4 Flux Summary

Metric	Value
Total observations	198.0
Rejections	
Rejected — poor model fit (n)	3.0
Rejected — model (%)	1.5
Rejected - [CH4] (n)	6.0
Rejected — [CH4] (%)	3.0
Model performance	
Passed — Linear model (n)	194.0
Passed — HM81 model (n)	120.0
Passed — Polynomial model (n)	195.0



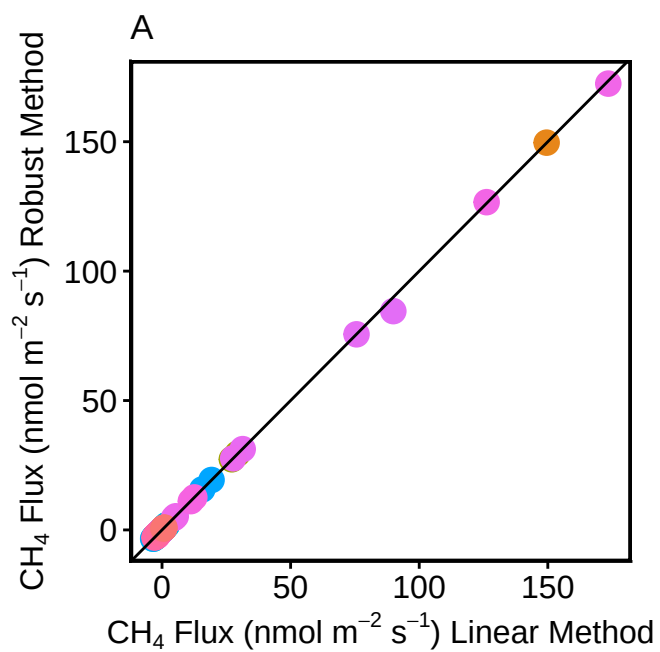
1.4 Remove any fluxes that do not fit criteria

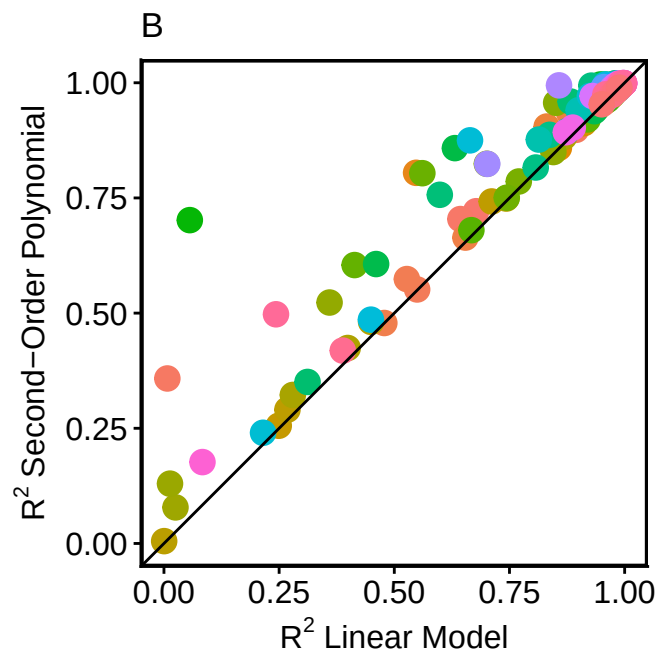
	Plot_ID	TIMESTAMP	HM81_AIC
1	2024_02_02_CRC_TR_36_1	2024-02-02 10:14:53	NA
2	2024_02_02_CRC_TR_36_2	2024-02-02 10:16:19	NA

3	2024_02_02_CRC_TR_36_B_1	2024-02-02 10:26:26	NA	
4	2024_02_02_CRC_TR_36_CORRECT_1	2024-02-02 10:31:01	NA	
5	2024_02_02_CRC_TR_36_CORRECTC_1	2024-02-02 10:35:26	NA	
6	2024_02_02_CRC_TR_36_CORRECTC_2	2024-02-02 10:36:51	NA	
	HM81_flux.estimate	HM81_p.value	HM81_r.squared	HM81_RMSE lin_AIC
1	NA	NA	NA	NA 125.2098
2	NA	NA	NA	NA 114.7222
3	NA	NA	NA	NA 129.1205
4	NA	NA	NA	NA 128.4288
5	NA	NA	NA	NA 113.3847
6	NA	NA	NA	NA 127.0208
	lin_flux.estimate	lin_flux.std.error	lin_int.estimate	lin_int.std.error
1	-0.080071128	0.007972073	17046.96	0.2618071
2	-0.066812059	0.007178342	17026.08	0.2357405
3	0.004939926	0.008290018	16997.18	0.2722485
4	-0.165974162	0.008232869	17003.62	0.2703717
5	-0.054209800	0.007082976	17013.63	0.2326086
6	-0.059397689	0.008117762	16999.44	0.2665916
	lin_p.value	lin_r.squared	lin_RMSE	poly_AIC poly_r.squared poly_RMSE
1	2.193700e-13	0.677594843	0.8134836	121.9478 0.7211790 0.7727735
2	2.515332e-12	0.643463663	0.7324899	109.3965 0.7041293 0.6816201
3	5.540494e-01	0.007343242	0.8459271	111.3221 0.3581111 0.6948724
4	4.497950e-25	0.894371607	0.8400956	130.7711 0.8978163 0.8440554
5	7.380296e-10	0.549619694	0.7227586	117.1930 0.5513434 0.7368893
6	2.405528e-09	0.527273027	0.8283499	125.8261 0.5739209 0.8033327
	rob_AIC	rob_converged	rob_flux.estimate	rob_flux.std.error rob_RMSE
1	125.2106	TRUE	-0.079974992	0.008506063 0.8144742
2	114.7287	TRUE	-0.066435539	0.007779566 0.7453685
3	129.4130	TRUE	0.006165937	0.008900450 0.6898134
4	128.4458	TRUE	-0.166816110	0.008293662 0.8438627
5	113.7744	TRUE	-0.051472200	0.007144119 0.7535298
6	127.0794	TRUE	-0.060758532	0.007797324 0.9372916
	TIMESTAMP_min	TIMESTAMP_max	soilp_c	soilp_m soilp_t CH4_initial
1	2024-02-02 10:14:29	2024-02-02 10:15:18	0.001	0 2.900000 2069.28
2	2024-02-02 10:15:55	2024-02-02 10:16:44	0.001	0 2.900000 2066.59
3	2024-02-02 10:26:02	2024-02-02 10:26:51	0.001	0 3.000000 2066.82
4	2024-02-02 10:30:37	2024-02-02 10:31:26	0.001	0 3.056364 2067.40
5	2024-02-02 10:35:02	2024-02-02 10:35:51	0.001	0 3.100000 2069.55
6	2024-02-02 10:36:27	2024-02-02 10:37:16	0.001	0 3.200000 2068.26
	CH4_lin_pass	CH4_HM81_pass	CH4_poly_pass	CH4_init_conc CH4_rejected_model
1	TRUE	FALSE	TRUE	FALSE FALSE
2	TRUE	FALSE	TRUE	FALSE FALSE
3	TRUE	FALSE	TRUE	FALSE FALSE

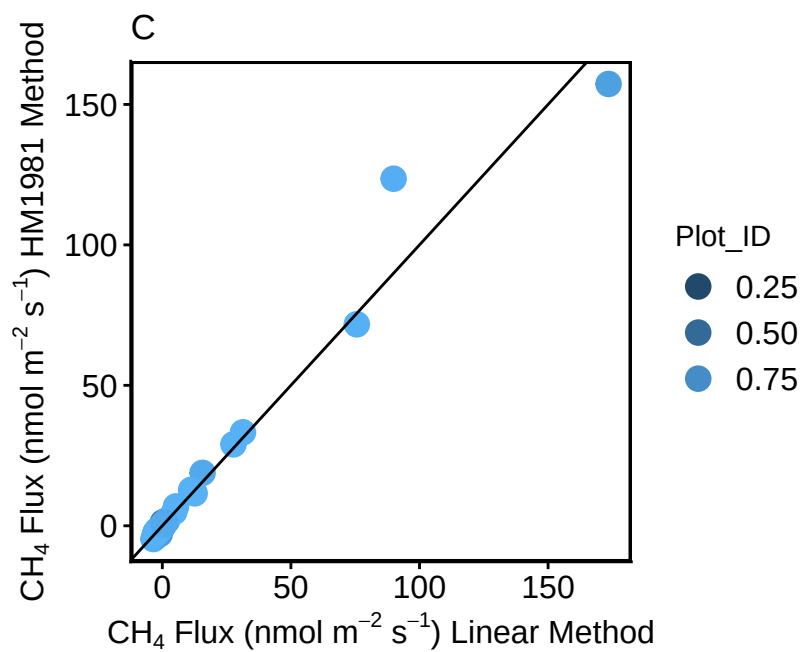
4	TRUE	FALSE	TRUE	FALSE	FALSE
5	TRUE	FALSE	TRUE	FALSE	FALSE
6	TRUE	FALSE	TRUE	FALSE	FALSE
	CH4_rejected_manual	CH4_flux_rejected			
1	FALSE	FALSE			
2	FALSE	FALSE			
3	FALSE	FALSE			
4	FALSE	FALSE			
5	FALSE	FALSE			
6	FALSE	FALSE			

1.5 Do some visualization of QAQC plots to assess fluxes





Warning: Removed 75 rows containing missing values or values outside the scale range (`geom_point()`).



1.6 Read out final dataframe and export for use elsewhere

```
[1] "TR" "UP" "W"
```

```
[1] 36 37 39 40 48 25 26 27 29 30 41 42 43 47 58 60 61 62 63 NA 50 51 52 55 56  
[26] 65 66 67 71 72 33 34 38 28 31 32 44 57 59 64 49 53 54 68 11 12 13 14  2  3  
[51]  4  6  7  8 18 19 21 22 23
```

```
[1] 1 2
```